

# Dakai Kang

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## Education

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**University of California, Davis**  
Computer Science, Ph.D.

Sep 2022 – Jun 2026  
Davis, CA

### Research Team

Exploratory Systems Lab ([expolab.org](https://expolab.org)), advised by Prof. Mohammad Sadoghi ([msadoghi@ucdavis.edu](mailto:msadoghi@ucdavis.edu)).

### Research Topics

- High-Performance Blockchain Infrastructure
- BFT Consensus Protocols; Transaction Scheduling; Parallel Execution; MEV Problem

### Teaching

- ECS165A, Databases, UC Davis, [Winter 2023](#), [2026](#) – Teaching Assistant
- ECS189F, Distributed Ledgers, UC Davis, [Fall 2023](#) – Teaching Assistant
- ECS265, Distributed Database Systems, UC Davis, [Fall 2024](#), [2025](#) – Teaching Assistant
- ECS188, Ethics in Computer Science, UC Davis, [Spring 2025](#) – Teaching Assistant

### Awards

- GGCS Spring Quarter Fellowship Award, UC Davis (2023, 2024)
- GGCS Summer Ph.D. Fellowship (2023, 2024)

**University of California, Davis**  
Computer Science, M.S.

Dec 2025  
Davis, CA

**University of California, Davis**  
GREAT (Global Research Experience in Advanced Technology) Program

Sep 2021 – July 2022  
Remote

Research Assistant Intern, Exploratory Systems Lab ([expolab.org](https://expolab.org)).

**Zhejiang University**  
Software Engineering, Bachelor

Sep 2018 – July 2022  
Hangzhou, China

### Awards

- The Third Prize Scholarship of Zhejiang University (2019, 2020, 2021)
- Excellent Engineer Scholarship of Software Engineering Major (2022)

## Work Experience

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**Sei Labs, Inc.** (<https://www.sei.io>)  
Research Scientist

Jun 2026 – Present

**Mysten Labs, Inc.** (<https://www.mystenlabs.com>)  
Research Intern

Jun 2025 – Sep 2025  
Remote

- Worked on a research project on transaction scheduling in Multi-proposer consensus to accelerate parallel transaction execution and mitigate congestion under high-contention workloads.

Hangzhou Zhuxing Information Technology Co., Ltd.

Fullstack Software Engineer Intern

Jul 2020 – Oct 2020

Hangzhou, China

## Publications

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- [1] Dinh, T. T. A., **Kang, D.**, & Sadoghi, M. (2026, July). FEAT: Fair and Efficient Adversarial Transaction Ordering. In *Proceedings of the 2026 ACM Symposium on Principles of Distributed Computing (PODC '26)*, Egham, United Kingdom. ACM. <https://doi.org/10.1145/3796701.3815905>
- [2] Xie, S., **Kang, D.**, Lyu, H., Niu, J., & Sadoghi, M. (2026). Fides: Scalable Censorship-Resistant DAG Consensus via Trusted Components. *Proceedings of the VLDB Endowment (VLDB 2026)*. (*To appear.*) <https://arxiv.org/abs/2501.01062>
- [3] **Kang, D.**, Chen, J., Dinh, T. T. A., & Sadoghi, M. (2025). FairDAG: consensus fairness over multi-proposer causal design. *Proceedings of the VLDB Endowment*, 19(2), 265–278.
- [4] Gupta, S., **Kang, D.**, Malkhi, D., & Sadoghi, M. (2025). Brief announcement: Carry the tail in consensus protocols. In *Proceedings of the 39th International Symposium on Distributed Computing (DISC 2025)* (Leibniz International Proceedings in Informatics [LIPIcs], Vol. 356, pp. 59:1–59:7). Schloss Dagstuhl – Leibniz-Zentrum für Informatik. <https://doi.org/10.4230/LIPIcs.DISC.2025.59>
- [5] **Kang, D.**, Gupta, S., Malkhi, D., & Sadoghi, M. (2025). Hotstuff-1: Linear consensus with one-phase speculation. *Proceedings of the ACM on Management of Data*, 3(3), 1–29. (*Honorable Mention for Best Artifact*)
- [6] **Kang, D.**, Rahnema, S., Hellings, J., & Sadoghi, M. (2024, May). Spotless: Concurrent rotational consensus made practical through rapid view synchronization. In *2024 IEEE 40th International Conference on Data Engineering (ICDE)* (pp. 1916–1929). IEEE.

## Preprints

- [7] Gupta, S., **Kang, D.**, Malkhi, D., & Sadoghi, M. (2025). Carry the Tail in Consensus Protocols. *arXiv preprint arXiv:2508.12173*.

## Open-Source Projects

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- [1] ResilientDB (under Apache Incubation, <https://expolab.resilientdb.com>), **Architect**
  - Github Repository: <https://github.com/apache/incubator-resilientdb>
  - Implemented BFT consensus protocols in ResilientDB, including PBFT, HotStuff-1 and FairDAG, etc.

## Peer-Review Service

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### Program Committee Member

- International Symposium on Reliable Distributed Systems (**SRDS**) 2026

### Invited Reviewer

- IEEE International Conference on Data Engineering (**ICDE**) (Demo Track) 2026
- IEEE Transactions on Information Forensics and Security (**T-IFS**) 2025
- IEEE Transactions on Dependable and Secure Computing (**TDSC**) 2025–2026
- ACM **SIGMOD** ARI (Availability & Reproducibility Initiative) 2025
- European Conference on Computer Systems (**EuroSys**) Shadow PC program 2026

### External Reviewer

- ACM Conference on Computer and Communications Security (**CCS**) 2025
- International Conference on Very Large Data Bases (**VLDB**) 2023–2025
- ACM Special Interest Group on Management of Data (**SIGMOD**) 2024–2025
- IEEE International Conference on Data Engineering (**ICDE**) 2024
- ACM/IFIP International Middleware Conference (**Middleware**) 2024